

National Occupational Standards



Green Logistics Practices

Unit Code: SGJ/N4601

Version: 1.0

NSQF Level: 4

Skill Council for Green Jobs || 3rd Floor, CBIP Building, Malcha Marg, Chanakyapuri
New Delhi - 110021

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Description

This standalone NOS is designed to incorporate environmentally sustainable practices and principles into logistics education and training. This aims to train existing and future professionals in the field of logistics while emphasizing the importance of reducing environmental impacts, optimizing resources, and promoting sustainable operations within the supply chain. Adopting green skills in logistics sector not only benefits the environment but can also lead to reduced operating costs, improved public perception, and compliance with evolving environmental regulations. It is essential for logistics operators to stay updated on the latest sustainable practices and technologies to remain competitive and environmentally responsible in the logistics and supply chain industry. In that context, the proposed NOS is to introduce generic green skills under logistics sector courses by dove-tailing green skill modules to the current and future logistics course curricula.

Scope

The scope covers the following :

- Promoting Sustainability through Green Logistics.

Elements and Performance Criteria

Basics of Climate Change, Environmental Concerns and Renewable Energy

To be competent, the user/individual on the job must be able to:

- PC1.** define Sustainable Development
- PC2.** explain UN Sustainable Development Goals (SDGs) with a focus on SDG 7 & 13
- PC3.** explain Green House Gas emissions and illustrate through examples the sources of GHG emissions in Logistics sector
- PC4.** discuss briefly the environmental Impacts of Logistics & and ways to address that
- PC5.** discuss the difference between the conventional and renewable energy sources
- PC6.** explain and illustrate the possible use of renewable energy in logistics sector
- PC7.** discuss how to estimate the electrical load of workplace and how to source the energy through renewable technologies
- PC8.** explain the concept of circular economy
- PC9.** illustrate a case study on circular economy in logistics sector
- PC10.** illustrate the importance of Green skills in logistics sector

Introduction to Green Logistics

To be competent, the user/individual on the job must be able to:

- PC11.** explain green logistics
- PC12.** explain and outline the differences between conventional logistics and green logistics
- PC13.** explain the drivers and key levers for green logistics in Indian context
- PC14.** discuss green logistics initiatives in India and illustrate a case study to list typical differences with conventional logistics
- PC15.** discuss and illustrate key Supply Chain Management practices and their impacts on environment

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- PC16.** discuss and illustrate Warehouse Management practices and their impact on environment
- PC17.** discuss and illustrate transport / shipping practices and their impact on environment
- PC18.** explain the components of Green Supply Chain Performance Metrics
- PC19.** outline the components of measuring Carbon footprint of a shipping / transport agency
- PC20.** list out examples of greener supply chain management
- PC21.** prepare a list of key measures which can be incorporated for greening the logistics sector

Steps required for Greening the Logistics Sector

To be competent, the user/individual on the job must be able to:

- PC22.** explain the Practices and Strategies to Green the Supply Chain Management
- PC23.** explain what are Sustainable and low carbon material and processes
- PC24.** explain how to Streamline inventory management for a greener supply chain
- PC25.** explain Resource Conservation (e.g., material, energy, water usage) and show how to manage different types of waste effectively
- PC26.** discuss and demonstrate how to Implement Energy/ Electricity/material/water Conservation Practices
- PC27.** explain the use of Renewable Energy with a focus on solar energy and its potential applications across the logistics sector
- PC28.** explain and show how to manage different types of Waste effectively
- PC29.** Discuss how to examine logistics operations focusing on energy, material, and waste management, while exploring successful cases of companies undertaking initiatives to promote environmentally sustainable logistics practices
- PC30.** explain the areas where concept of Circular economy be introduced
- PC31.** how can efficient green logistics be implemented and list and suggest green practices at the work area
- PC32.** discuss on the importance of optimizing logistics operations to achieve sustainability goals
- PC33.** outline measures to make workplace operations/activities environment friendly

Identifying Green Supply and Demand networks

To be competent, the user/individual on the job must be able to:

- PC34.** explain Green Purchasing and its Importance
- PC35.** explain the ways for Green Purchasing in Supply Chain Management
- PC36.** discuss and identify the best Practices for Green Purchasing
- PC37.** explain and outline the criteria for selecting a supplier with green attributes
- PC38.** discuss the supplier evaluation , selection based on green approach in Supply Chain Management and explain how to choose a green logistics partner
- PC39.** Illustrate how to enable the transporter to adopt green practices
- PC40.** prepare the Map of entire supply chain, identifying suppliers, transportation routes, manufacturing processes, and end consumers and outline on reducing the environmental impacts at each stage
- PC41.** prepare an assessment criterion to evaluate suppliers based on their sustainability practices, covering aspects like energy efficiency, waste management, and ethical sourcing
- PC42.** illustrate the case studies of logistics companies using upgraded technologies/Green practices and how it benefitted the organisation

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- PC43.** demonstrate the implementation of smart logistics solutions, through pictures or videos
- PC44.** show emerging technologies like autonomous vehicles and their potential applications in green logistics through case studies, pictures or videos

Health and Safety in Green Logistics

To be competent, the user/individual on the job must be able to:

- PC45.** explain Occupational Health & Safety Standards and Regulations along with the requirements for safe work area in Green Logistics
- PC46.** discuss about potential emergency situations in various operations across Green Logistics
- PC47.** explain and show the process of Administering first aid
- PC48.** explain Work Safety Procedures and Instructions in Green Logistics
- PC49.** discuss about Fire Protection in Green Logistics and demonstrate the use of fire extinguisher
- PC50.** explain and demonstrate about use of PPEs used in Green Logistics
- PC51.** discuss and demonstrate about Housekeeping and Infection Control guidelines
- PC52.** illustrate a case study of managing safety in Green Logistics sector

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational procedures and policy on procurement
- KU2.** company's reporting structure to support and expedite project activities
- KU3.** company's policy and work instructions on quality standards
- KU4.** policy on return of goods
- KU5.** importance of the individual's role in the workflow
- KU6.** occupational health and safety standards and security procedures to be followed
- KU7.** procedures for dealing with loss or damaged goods
- KU8.** risk and impact of not following defined work, safety and security procedures
- KU9.** Enterprise Resource Planning (ERP) system of the organization
- KU10.** ability to understand the engineering drawings and product specifications
- KU11.** use of computer and data handling devices
- KU12.** technical specification of products handled
- KU13.** packaging for different types of products handled
- KU14.** various escalations regarding resolving and catering to the customer query
- KU15.** overall procurement process
- KU16.** handling of quality issues and rejects

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read order requirement
- GS2.** read invoices

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- GS3.** write damage reports and daily output reports
- GS4.** write Material Return Note (MRN)
- GS5.** decide if a problem can be resolved quickly internally or needs to be escalated
- GS6.** identify activities or orders that need to be prioritised as per instructions
- GS7.** decide as per the organisation guiding policies and criteria regarding procurement
- GS8.** plan and estimate the time for each activity
- GS9.** respond to the supervisor in a timely manner
- GS10.** prioritize and execute tasks based on instructions
- GS11.** manage timelines
- GS12.** inform the supervisor regarding delays, issue in doing an activity, etc.
- GS13.** focus on task at hand and complete it without errors and delays

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Basics of Climate Change, Environmental Concerns and Renewable Energy</i>	9	6	-	-
PC1. define Sustainable Development	1	-	-	-
PC2. explain UN Sustainable Development Goals (SDGs) with a focus on SDG 7 & 13	1	-	-	-
PC3. explain Green House Gas emissions and illustrate through examples the sources of GHG emissions in Logistics sector	1	2	-	-
PC4. discuss briefly the environmental Impacts of Logistics & and ways to address that	1	-	-	-
PC5. discuss the difference between the conventional and renewable energy sources	1	-	-	-
PC6. explain and illustrate the possible use of renewable energy in logistics sector	1	-	-	-
PC7. discuss how to estimate the electrical load of workplace and how to source the energy through renewable technologies	2	-	-	-
PC8. explain the concept of circular economy	1	-	-	-
PC9. illustrate a case study on circular economy in logistics sector	-	2	-	-
PC10. illustrate the importance of Green skills in logistics sector	-	2	-	-
<i>Introduction to Green Logistics</i>	9	16	-	-
PC11. explain green logistics	1	-	-	-
PC12. explain and outline the differences between conventional logistics and green logistics	1	2	-	-
PC13. explain the drivers and key levers for green logistics in Indian context	1	-	-	-
PC14. discuss green logistics initiatives in India and illustrate a case study to list typical differences with conventional logistics	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. discuss and illustrate key Supply Chain Management practices and their impacts on environment	1	2	-	-
PC16. discuss and illustrate Warehouse Management practices and their impact on environment	1	2	-	-
PC17. discuss and illustrate transport / shipping practices and their impact on environment	1	2	-	-
PC18. explain the components of Green Supply Chain Performance Metrics	1	-	-	-
PC19. outline the components of measuring Carbon footprint of a shipping / transport agency	1	-	-	-
PC20. list out examples of greener supply chain management	-	2	-	-
PC21. prepare a list of key measures which can be incorporated for greening the logistics sector	-	4	-	-
<i>Steps required for Greening the Logistics Sector</i>	16	18	-	-
PC22. explain the Practices and Strategies to Green the Supply Chain Management	1	-	-	-
PC23. explain what are Sustainable and low carbon material and processes	1	-	-	-
PC24. explain how to Streamline inventory management for a greener supply chain	1	-	-	-
PC25. explain Resource Conservation (e.g., material, energy, water usage) and show how to manage different types of waste effectively	1	2	-	-
PC26. discuss and demonstrate how to Implement Energy/ Electricity/material/water Conservation Practices	1	3	-	-
PC27. explain the use of Renewable Energy with a focus on solar energy and its potential applications across the logistics sector	2	-	-	-
PC28. explain and show how to manage different types of Waste effectively	1	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC29. Discuss how to examine logistics operations focusing on energy, material, and waste management, while exploring successful cases of companies undertaking initiatives to promote environmentally sustainable logistics practices	2	3	-	-
PC30. explain the areas where concept of Circular economy be introduced	2	-	-	-
PC31. how can efficient green logistics be implemented and list and suggest green practices at the work area	2	3	-	-
PC32. discuss on the importance of optimizing logistics operations to achieve sustainability goals	2	2	-	-
PC33. outline measures to make workplace operations/activities environment friendly	-	2	-	-
<i>Identifying Green Supply and Demand networks</i>	7	8	-	-
PC34. explain Green Purchasing and its Importance	1	-	-	-
PC35. explain the ways for Green Purchasing in Supply Chain Management	1	-	-	-
PC36. discuss and identify the best Practices for Green Purchasing	1	-	-	-
PC37. explain and outline the criteria for selecting a supplier with green attributes	1	-	-	-
PC38. discuss the supplier evaluation , selection based on green approach in Supply Chain Management and explain how to choose a green logistics partner	1	-	-	-
PC39. Illustrate how to enable the transporter to adopt green practices	1	-	-	-
PC40. prepare the Map of entire supply chain, identifying suppliers, transportation routes, manufacturing processes, and end consumers and outline on reducing the environmental impacts at each stage	1	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC41. prepare an assessment criterion to evaluate suppliers based on their sustainability practices, covering aspects like energy efficiency, waste management, and ethical sourcing	-	1	-	-
PC42. illustrate the case studies of logistics companies using upgraded technologies/Green practices and how it benefitted the organisation	-	2	-	-
PC43. demonstrate the implementation of smart logistics solutions, through pictures or videos	-	2	-	-
PC44. show emerging technologies like autonomous vehicles and their potential applications in green logistics through case studies, pictures or videos	-	2	-	-
<i>Health and Safety in Green Logistics</i>	7	4	-	-
PC45. explain Occupational Health & Safety Standards and Regulations along with the requirements for safe work area in Green Logistics	1	-	-	-
PC46. discuss about potential emergency situations in various operations across Green Logistics	1	-	-	-
PC47. explain and show the process of Administering first aid	1	1	-	-
PC48. explain Work Safety Procedures and Instructions in Green Logistics	1	-	-	-
PC49. discuss about Fire Protection in Green Logistics and demonstrate the use of fire extinguisher	1	-	-	-
PC50. explain and demonstrate about use of PPEs used in Green Logistics	1	1	-	-
PC51. discuss and demonstrate about Housekeeping and Infection Control guidelines	1	1	-	-
PC52. illustrate a case study of managing safety in Green Logistics sector	-	1	-	-
NOS Total	48	52	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4601
NOS Name	Green Logistics Practices
Sector	Green Jobs
Sub-Sector	Other Green Jobs
Occupation	Green Logistics
NSQF Level	4
Credits	1
Minimum Job Entry Age	NA
Minimum Educational Qualification & Experience	12th Class (or equivalent) OR 10th Class with 2 Years of experience OR 8th grade pass plus 2-year NTC plus 1 Year NAC with 2 Years of experience
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQF Clearance Date	31/01/2024
Reference code on NQR	NG-04-ES-02095-2024-V1-SCGJ
NQR Version	1
CCN Category	2